Wi-Fi Direct Research - Current Status and Future Perspectives

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Abstract

Wi-Fi Direct, also named Wi-Fi P2P, is a popular technology that enables device to device communication in Wireless LAN. It allows Wi-Fi Direct-enabled devices to dynamically negotiate and select one of the mobile devices as Group Owner. The Group Owner plays the role of an Access Point as in Wi-Fi infrastructure mode. Wi-Fi Direct protocol was initially released to connect Wi-Fi enabled devices on the fly. However, due to the enhanced features, the protocol has the potential to be used in several applications such as files transferring, sharing resources, online gaming, alert dissemination, social networking, etc. An extensive survey on Wi-Fi Direct technology and a review of state-of-the-art is presented in this paper. An insight into current research works, applications, and challenges related to Wi-Fi Direct is provided. Moreover, the paper presents our proposed Group Formation scheme, which aims to enhance the performance of Wi-Fi Direct networks.

Keywords: Wi-Fi Direct, Peer-to-Peer, Access Point, Group Owner, Network Simulators

1. Introduction

Wi-Fi has become a choice for short range communication in many applications, due to the cost effective deployment of Wi-Fi networks, its large scale implementation and availability of Wi-Fi devices e.g. smartphones, consumer electronics and industry sensors. In the current deployment of Wi-Fi infrastructure mode, devices connect to a common Access Point (AP) to connect to other Wi-Fi devices.

Wi-Fi Alliance [1] introduced in 2010 the Wi-Fi Direct technology to enable Wi-Fi devices to directly connect to each other without connecting to an AP. Wi-Fi Direct, initially called Wi-Fi Peer-to-Peer (Wi-Fi P2P), is built upon the IEEE 802.11 *infrastructure* mode and offers direct, secure and rapid device to device communication. The recent Wi-Fi P2P Technical Specification [1] was released in 2016 (version 1.7). Wi-Fi Direct enabled devices discover each other and form a P2P Group. In each P2P Group, one elected node, called P2P Group Owner, acts as AP.

The Wi-Fi Direct is becoming an interesting and suitable candidate for communication in several application domains including content distribution, resource sharing, emergency communication, alert dissemination, online gaming, proximity based advertising and social networking.

The contribution of this article is to present a review of research efforts on Wi-Fi Direct in three main areas: performance evaluation, technology improvement and applications development. The paper also includes an enhanced modified Group Formation scheme to reduce Group Formation delay.

The remainder of the paper is as follow: Section 2 presents an overview of Wi-Fi Direct technology including network architecture, Device and Service Discovery mechanisms, Group Formation and power saving schemes. Section 4 covers state-of-the-art in Wi-Fi Direct research focused on performance evaluations and enhancements of various features of Wi-Fi Direct technology. Section 5 provides a review of literature on the applications of Wi-Fi Direct. Section 6 provides a short review of relay and leader selection algorithms to evaluate their applicability for the Group Owner selection in Wi-Fi Direct. Section 7 presents a modified Group Formation scheme to reduce the delay encountered at various phases of forming a P2P Group. Section 8 highlights the key challenges to Wi-Fi Direct development and significant opportunities. To the best of our knowledge, this is the first article that covers every aspect in Wi-Fi Direct research space.

2. Technical overview

Wi-Fi Direct enables Wi-Fi devices such as smartphones, laptops, smart TVs, printers, cameras and other appliances to inter-connect quickly and conveniently without incorporating an Access Point (AP). Wi-Fi Direct is built on the *infrastructure* mode of WLAN. Wi-Fi Direct connections are secured with Wireless Protected Access - 2 (WPA2) [2]. Wi-Fi Direct supports the same high data rates as in Wi-Fi (up-to 250 Mbps). The range of Wi-Fi Direct connection is 200 meters (this is theoretical range and practical range might be nominal of this). The specifications also require 1:1 connection mandatory for Wi-Fi Direct certified devices, whereas keeping 1:N connection optional feature.

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[1] http://www.wi-fi.org/

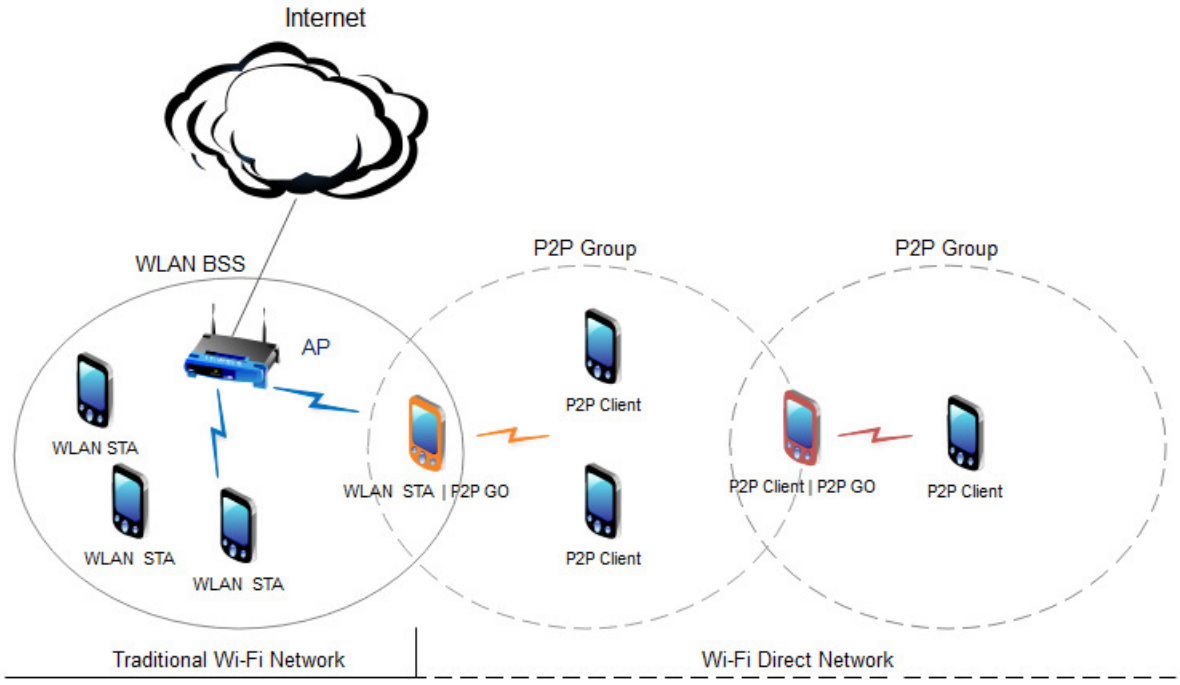

Figure 1: Wi-Fi and Wi-Fi Direct Network Architectures.

In the subsequent sections, we provide a detailed overview of Wi-Fi Direct features.

2.1. Network Architecture

The functional entity of Wi-Fi Direct architecture is called a "P2P Group" that is functionally equivalent to a Basic Service Set (BSS) in *legacy* Wi-Fi network. A P2P Group consists of a P2P Group Owner (P2P GO) and zero or more P2P Clients. The P2P GO (sometimes referred to as "GO") is also called a *Soft-AP*. AP functions are implemented within Wi-Fi P2P devices. A P2P device can dynamically take the role of an AP or client. The roles of P2P Devices (i.e. P2P GO and P2P Client) are usually negotiated before creating a P2P Group and remain fixed while the P2P Group is active. Figure 1 illustrates the different roles of P2P Devices.

2.2. Device Discovery

Device Discovery is a mandatory feature to be supported by all P2P Devices. Prior to form a P2P Group, a P2P Device runs the Device Discovery procedure to detect the presence of other P2P Devices in its wireless range. The procedure consists of two distinct phases: *Scan* and *Find*. In the *Scan* phase, the P2P Device performs traditional Wi-Fi scan (passive scan) through all supported channels in order to collect information about the surrounding devices, P2P Groups and legacy Wi-Fi networks. Once the *Scan* phase is completed, the device enters into the *Find* phase. In the *Find* phase, the P2P Device alternates between two states: *Search* and *Listen*. In the *Search* state, the P2P Device sends one or more *Probe Request (PREQ)* frames on the social channel namely channels 1, 6 and 11 in the 2.4 GHz band. In the *Listen* state, the P2P Device dwells on one of the social channels (1, 6 and 11) called the *Listen* channel and waits for *Probe Request (PREQ)* frames from other P2P Devices. Thus, the success of the *Find* phase is that when two devices come to a common channel to communicate. It is noticeable that the P2P Device Discovery process can induce some delay to let a P2P Device discovers all P2P Devices in its vicinity. This delay, termed as "Device Discovery delay", can be relatively high if several P2P Devices are simultaneously performing Device Discovery in the same wireless range. Figure 2 illustrates the P2P Device Discovery procedure in Wi-Fi Direct.

2.3. Service Discovery

Service Discovery is an optional procedure in Wi-Fi Direct. The procedure starts after the Device Discovery and prior to the Group Formation procedure. It allows a P2P Device to connect to other P2P Devices only if the latter offers the intended service. Using Service Discovery procedure, a P2P Device advertises available services using link layer Generic Advertisement Service (GAS) protocol [3]. Wi-Fi Alliance has defined a set of standardized services supported by Wi-Fi Direct such as *Play*, *Send* and *Print* [4].

2.4. Group Formation

Following a successful Device Discovery (mandatory procedure) and Service Discovery (optional procedure), P2P Devices can establish the P2P Group. During the Group Formation, the device that will act as GO is determined. As described in Figure

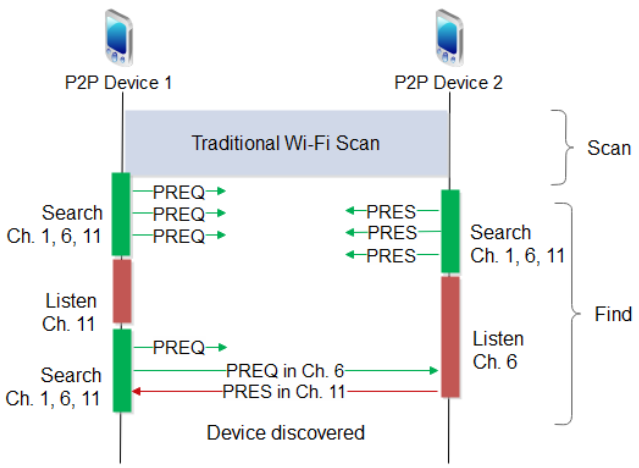

Figure 2: Device Discovery Procedure.

3, three types of P2P Group Formation schemes are possible in Wi-Fi Direct: (1) Standard Group Formation (2) Autonomous Group Formation and (3) Persistent Group Formation.

In *Standard* Group Formation, presented in Figure 3a, two P2P Devices negotiate the role of the P2P GO. The GO Negotiation is a three-way handshake. During the handshake, the two devices send to each other a randomly chosen numeric value called "Intent value". The Intent value ranges from 0 to 15, and it measures the desire of the P2P Device to be the P2P GO. The P2P Device sending the higher Intent value shall become GO. In case both P2P devices send equal GO Intent values, a tie breaker bit is used for decision and the device with tie breaker bit set to 1 shall become GO. Figure 4 illustrates the Intent value comparison between two P2P devices during Standard Group Formation.

The P2P Device selected as P2P GO shall start a P2P Group session. The other P2P Device can then connect to the P2P GO to obtain credentials, and exchange data. Similarly, other P2P Devices and legacy Wi-Fi devices can join the P2P Group as clients.

In *Autonomous* Group Formation, depicted in Figure 3b, the role of GO is not negotiated. Instead, a P2P Device announces itself as GO and starts sending Beacons. This process is much similar to the legacy Wi-Fi in which an AP directly sends Beacons into the network to become discoverable. The Autonomous Group Formation is simpler and faster than Standard Group Formation.

In *Persistent* Group Formation, depicted in Figure 3c, a P2P Device sends an invitation to another P2P Device, which was previously connected to it in a P2P Group, in order to re-instantiate the P2P Group. This is accomplished using *P2P Invitation Request* and *P2P Invitation Response* frames. The role of each P2P Device shall remain the same as in the previously formed P2P Group. To establish a Persistent group, the P2P Devices must declare the P2P Group as Persistent during the Standard or Autonomous formation of the group. A flag bit inside the P2P Beacons, Probe Response and GO Negotiation frames is used to indicate that the P2P Group is Persistent or not. If the flag is not set during Group Formation procedure, the P2P Devices cannot re-instantiate a Persistent group in future and must starts a Standard or Autonomous group.

The Wi-Fi Direct specification [1] defines the Standard and Persistent Group Formation procedures only between two P2P Devices. Other P2P Devices can only join, as clients, an already formed P2P Group.

2.5. *Power Saving Schemes*

Legacy Wi-Fi uses power saving scheme using *Sleep* and *Active* modes for Wi-Fi STAs (clients). Most of the traditional APs are permanently connected to a regular power source, and thus, they have no need for any power saving feature. However, in Wi-Fi Direct, the P2P GO, which acts as a Soft-AP, is a battery powered device and have limited lifetime. Hence, Wi-Fi Direct introduces two novel schemes for power saving in the P2P Devices. These schemes are: (1) Opportunistic Power Save (OppPS) and (2) Notice of Absence (NoA).

In OppPS scheme, the GO is allowed to save power when its clients are in the Sleep mode. The GO announces its presence period called "CTWindow". At the end of the CTWindow, if all nodes are in *Sleep* mode, the GO can also go to *Sleep* mode until the next Beacon. However, at the end of CTWindow, if one of the P2P Client nodes is in *Active* mode, then the GO must remain active until the next Beacon.

In NoA scheme, the GO announces via Beacons and Probe Response frames, an "absence period". During the absence period, its clients cannot access the channel, thus the GO shut down its radio to save energy used in transmission or reception. The absence period is announced in Beacons using NoA schedule, consisting of four parameters: 1) *Duration* - the length of absence period, 2) *Interval* - the time between two consecutive absence periods, 3) *Start Time* - the start time of the first absence period after the current Beacon, and 4) *Count* - the number of absence periods in the current NoA schedule. The Wi-Fi Direct specification does not define the values of these parameters.

2.6. *Security*

Wi-Fi Direct requires all P2P Devices to implement *Wi-Fi Protected Setup (WPS)* [5] in order to secure the connection establishment process and communication in the P2P Group. In WPS scheme, the P2P GO implements the internal *Registrar* whereas the P2P Client implements *Enrollee*. The WPS scheme works into two phases. In phase 1, the internal Registrar generates and issues the network credentials to Enrollee. In phase 2, the Enrollee (P2P Client) reconnects to the internal Registrar (P2P GO) using the new credentials.

3. Evaluation Tools

Wi-Fi Direct is already implemented and integrated in recent Android devices. Currently, researchers use prototyping on Android devices to evaluate their outcomes. Prototyping is not al-

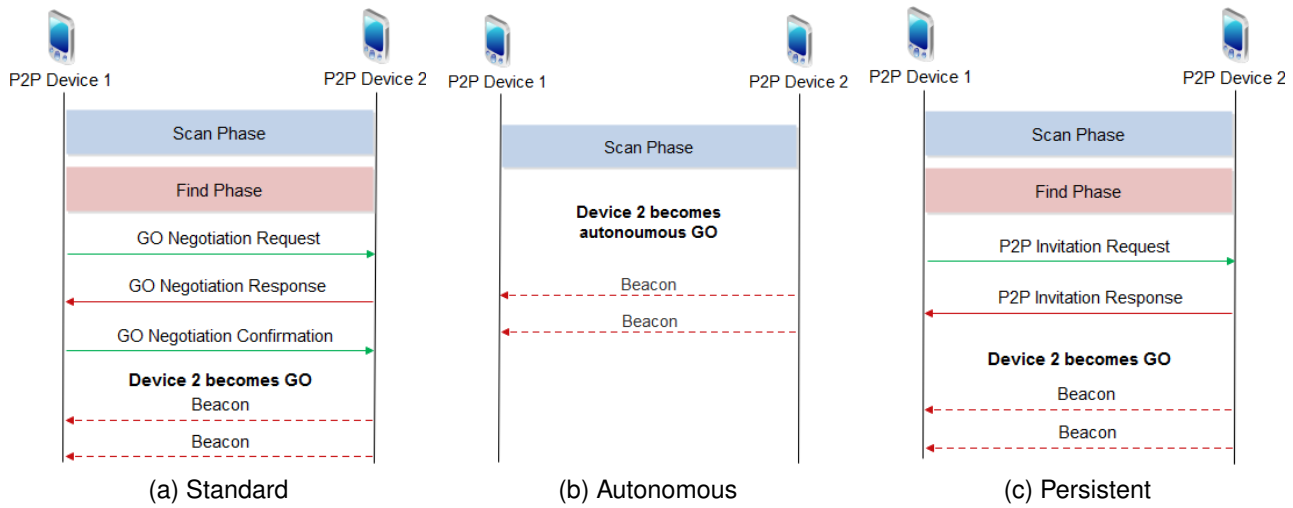

Figure 3: Wi-Fi Direct Group Formation methods.

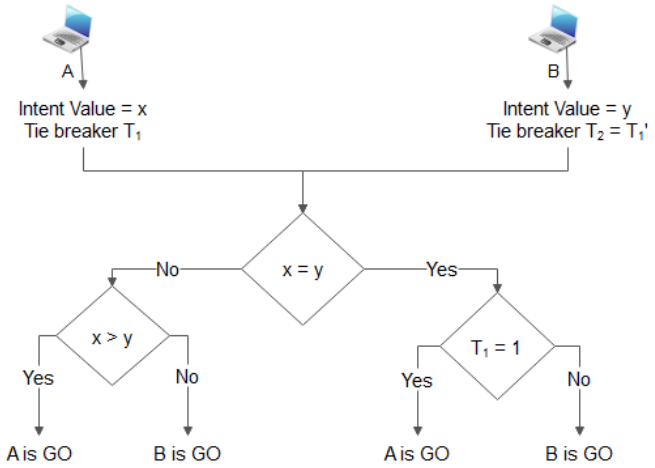

Figure 4: Intent Value Comparison.

ways suitable for complex scenarios evaluation. Therefore, researchers use simulators in order to evaluate the performances of the network in complex scenarios with a large number of nodes before prototyping.

This section presents a non exhaustive list of software and frameworks that can be used for simulating Wi-Fi Direct features.

3.1. Network Simulator - NS3

NS3[2] is a widely used network simulator in academic research. It is an open-source discrete-event network simulator. Several network protocols are already integrated in NS3: IP based wireless networks (such as Wi-Fi, WiMAX, LTE), non-IP based wireless networks (such as ZigBee) and even wired networks (such as Point-to-Point and CSMA).

For the time being, the main disadvantage of NS3 is the lack of support for some communication protocols such as Wi-Fi Direct. The P2P Device and Service Discovery procedures are not currently implemented in NS3. Other Wi-Fi Direct features, common to those of legacy Wi-Fi can still be simulated in NS3. Some papers such as [6] and [7] claim to use NS3 in their work on Wi-Fi Direct, however they did not provide any details about their implementation.

3.2. WDSim

WDSim is a framework developed, at the University of Lisbon[3], in order to simulate the Android Wi-Fi Direct API. WDSim framework can be used on virtual machines that run Android applications. The implementation of the framework is very basic, and it does not include all Wi-Fi Direct features such as Group Formation, Dynamic Host Configuration Protocol (DHCP) server.

3.3. WiDiSi

WiDiSi [8] is an open-source system-level simulator for Wi-Fi Direct networks. It is one of the first Wi-Fi Direct simulators for Android applications. It is designed to support a large number of Wi-Fi Direct-enabled nodes and user-defined mobility patterns. WiDiSi is written in Java and its source code is available on Github[4]. WiDiSi is built on top of PeerSim [9], an open-source simulation framework. It is capable of simulating large-scale peer to peer networks.

3.4. OMNET++

OMNeT++[5] is an extensible, modular, component-based C++ simulation library and framework. Its primary use is for

[2] https://www.nsnam.org/

[3] https://tecnico.ulisboa.pt/pt/

[4] https://github.com/nasser1941/WiDiSi

[5] https://omnetpp.org/

real-time simulation and network emulation. OMNeT++ provides a component architecture for models. Components (modules) are programmed in C++, then assembled into larger components and models using a high-level language, called NED. An open-source OMNeT++ model, named INET[6], is used for wired, wireless and mobile networks. Authors in [10] have implemented Wi-Fi Direct in the INET framework of OMNeT++. The implementation includes the main procedures of Wi-Fi Direct such as Device Discovery, GO Negotiation and Group Formation. The source code of the implementation is available online[7].

3.5. Mac80211 hw_sim

Mac80211_hwsim[8] is a Linux kernel driver. This module allows users to simulate number of Wi-Fi radios for the mac80211. It can be used to test most of the mac80211 functionality in a way that matches very closely with the normal case of using real WLAN hardware. It already includes Wi-Fi Direct implementation and can be used to simulate the Device and Service Discovery mechanisms and the P2P Group Formation procedure. The original implementation does not support mobility. A modified driver has been developed [11] so that mobility and the medium can be simulated. Authors in [12] have used Mac80211_hwsim to validate and simulate a new P2P Group Formation procedure.

3.6. Riverbed Modeler

Riverbed Modeler[9], formerly called OPNET Modeler, is a licensed network simulator tool. Several network protocols are available in Riverbed Modeler, but it does not include the Wi-Fi Direct technology. Some researchers use Riverbed Modeler to evaluate specific features of the Wi-Fi Direct. For example, authors in [13] have simulated in Riverbed Modeler the Wi-Fi Direct Opportunistic Power Save (OppPS) and the Notice of Absence (NoA) protocols.

3.7. Matlab

Matlab[10] (Matrix laboratory) is an interactive software system for numerical computations and graphics. Matlab does not integrate the Wi-Fi Direct technology, but it can be used to evaluate analytic models of specific features of Wi-Fi Direct, such as energy consumption and throughput of P2P Devices within a P2P Group.

4. Recent Developments in Wi-Fi Direct

Currently, researchers are working on the evaluation and enhancement of Wi-Fi Direct technology. The works focus on experimental evaluation of the protocol's features as well as the development of novel services using Wi-Fi Direct. The state-of-the-art in Wi-Fi Direct is presented in this section.

[6] https://inet.omnetpp.org/

[7] https://www-phare.lip6.fr/$\sim$trnguyen/research/wifidirect/

[8] https://wireless.wiki.kernel.org/en/users/drivers/mac80211_hwsim

[9] https://www.riverbed.com

[10] https://www.mathworks.com

4.1. Device and Service Discovery

In [7], authors proposed "Listen Channel Randomization (LCR)" scheme to reduce *Device Discovery* delay caused in the *Find* phase. The proposed scheme was implemented using Markov model and tested using NS3 simulation.

Authors in [6] presented performance evaluation of the Device Discovery and Group Formation delay using Standard, Persistent and Autonomous Group Formation mode. Furthermore, the authors investigated the power saving techniques of Wi-Fi Direct using NS3 and *wpa_supplicant* software.

4.2. Group Formation

Authors in [14] proposed multi-hop communication in Wi-Fi Direct using P2P Concurrent Device and Ad-hoc On Demand Distance Vector (AODV) reactive routing protocol [15]. Routing is established during Device Discovery phase to allow multi-hop communication. The paper did not evaluate the performance of the proposed scheme.

In [16], authors proposed EMC (Efficient Multi group formation and Communication) protocol for Wi-Fi Direct. In EMC scheme, the P2P GO is elected based on remaining battery status of the device. The proposed scheme was tested in Android API.

In [17], authors proposed a dynamic election of P2P GO in Wi-Fi Direct. The elected leaders can be replaced dynamically based on clustering strategy such as battery status, speed of the user and direction of motion etc. The authors also presented a template for writing clustering algorithms for efficient Group Formation.

Authors in [18] proposed a combined metric approach to select the P2P GO based on several parameters. The parameters are normalized and weighted to compute Intent Values of each device. An election algorithm is then used to select the P2P GO.

In [19], authors proposed an alteration to the standard Group Formation. In their proposed scheme, during GO Negotiation, each device sends an flag bit set to 1, to indicate its willingness to serve as the so called Emergency Group Owner (EGO). The EGO is present in the group after the GO is selected. The EGO takes over the group and assumes GO role if the GO leaves voluntarily or involuntarily. The EGO implementation reduce the duration of connection disruption due to GO unavailability.

4.3. Power Saving

In [6], authors presented two approaches to compute absence periods for energy saving using NoA scheme: A *static policy*, where the P2P GO advertises a fixed duration presence window after each Beacon frame and a *dynamic policy*, where the P2P GO adjusts the duration of the presence window based on the traffic conditions using Adaptive Single Presence Period (ASPP) algorithm.

In [20], a Traffic Aware Parameter Tuning Scheme (TAPS) is proposed to dynamically adjust the parameters for OppPS and NoA schemes to improve the performance of Wi-Fi Direct network. The results show that TAPS consumes less energy

when the maximum values for the OppPS and NoA parameters are used whereas no improvement in throughput. However TAPS improves the throughput when the lowest values of these parameters (in the range) are used, whereas more energy was consumed by TAPS.

The OppPS and NoA schemes are also considered insufficient in services which are using periodic data transmissions such as video streaming, screen sharing, multi-user on-line gaming and video surveillance as stated in [21]. The authors in[21] presented Dynamically Synchronized Power Management (DSPM) scheme to enhance the efficiency of OppPS and NoA schemes. The DSPM scheme dynamically synchronize the active time slots with the data transmission time slots to further enhance the energy efficiency of OppPS and NoA schemes.The DSPM scheme is evaluated with analytical modeling and NS3 simulations.

In [22], authors proposed a similar approach to enhance the energy efficiency of Wi-Fi Direct power saving schemes. The authors proposed an algorithm to uses the probability distribution function (PDF) of the variable bit rate (VBR) video frames and dynamically adjusts the length and number of awake intervals in each beacon interval. The proposed scheme was evaluated for a single P2P client associated with a P2P GO.

In [13], authors proposed two algorithms named as Adaptive Single Presence Period (ASPP) and Adaptive Multiple Presence Periods (AMPP) to improve the performance of OppPS and NoA schemes. The ASPP scheme can be used with both OppPS and NoA, whereas the AMPP scheme can be used only with NoA however AMPP can also improve the performance of ASPP. In ASPP scheme, the P2P GO compute a single presence period in each beacon interval based on the amount of traffic at the up-link of the P2P GO. The presence period is shortened if the up-link is congested and vice versa. In contrast to ASPP, the AMPP uses multiple presence periods in a beacon interval and hence it can be used only with NoA scheme. An algorithm running on P2P GO estimates the raw bandwidth on the up-link external network and accordingly adjust the number of presence periods.

In [21], authors proposed Dynamic Power Save (DPS) scheme. A P2P GO which implements DPS scheme, toggles between OppPS and NoA schemes based on the traffic characteristics. The authors evaluated using simulations performed in NS3 that OppPS scheme outperform in bulky data transmissions while NoA outperform in streaming applications. Therefore the P2P GO must be capable to decide and switch the power save mode if the applications in the network are changed.

4.4. Extension, Integration and Evaluation

Several works proposed either extension of existing features of Wi-Fi Direct protocol or integration with other protocols and algorithms for performance improvements and realization of new services. In [23] a "Decentralized Collaborative Media Content Streaming" (named DOMS) system is proposed to realize flexible and scalable media content sharing among Wi-Fi devices. The authors used iDLS (inter-BSS Direct Link Setup) with Wi-Fi Direct to enable Wi-Fi devices to connect to each other. Each device sequentially download video contents from a set of sources and re-distribute the contents. A centralized scheduling algorithm is implemented by DOMS to distribute the video segments.

In [24], authors implemented multi-hop networking using Wi-Fi Direct. The authors proposed multi-hop networks over "Content Centric Network" (CCN) with Wi-Fi Direct to realize the content storage over all Wi-Fi devices. Content Centric Networking supports network caching by storing identical contents between source and destination, in order to reduce bandwidth consumption and content retrieval time. The P2P Concurrent Device is used as relay node.

In [25], authors discuss the problem of reliability in multicast communication over Wi-Fi Direct. The authors evaluated the performance of standard and leader based multicast for Wi-Fi Direct and proposed an "Enhanced Leader Based Multicast" (ELBM). In ELBM scheme, GO chooses one of its client as *Representative* client, called CTS-R for sending CTS (Clear to Send) frames. GO sends RTS (Request to Send) frame to CTS-R, which replies with CTS and then GO starts sending multicast traffic. The selection of CTS-R is proposed as the closest node to the P2P GO in order to minimize the interference and maximize the coverage.

In [26], the performance evaluation of Wi-Fi Direct for multimedia delivery and streaming applications was presented. The Round Trip Time (RTT) and bandwidth were chosen as performance metrics. Results, acquired from experiments using real test-bed, were measured and compared in Wi-Fi Direct versus in Wi-Fi networks. In addition, the impact of increasing number of P2P Devices were studied.

In [27], authors developed a Wi-Fi Direct based data transfer system called "Electrical Business Card Communication over Wi-Fi Direct (EriCC-W)" to identify smartphones using gestures and transforms the gestures into frequency spectrum to authenticate devices. The proposed scheme was tested and results were compared with the same system over Bluetooth. The evaluated performance metrics were connection establishment overhead and delay.

In [28], authors proposed a Wi-Fi Direct based "Voluntary Advertisement Dissemination Algorithm (VADA)" for e-commerce applications to transmit local advertisements to many users with low economical cost.

The above-mentioned work on Wi-Fi Direct is summarized in Table 1.

5. Applications of Wi-Fi Direct

Wi-Fi Direct is an emerging technology and has many interesting applications in home and office networking, multimedia content delivery, emergency communication, e-commerce, delay tolerant networks and on-line gaming. In this section, different use cases of Wi-Fi Direct are presented and summarized in Table 2. A list of applications using Wi-Fi Direct available on Android App Store[11] are provided in Table 3.

[11] https://play.google.com/store

Table 1: State of the Art in Wi-Fi Direct

Ref.	Research Area	Contribution	Tools	Details
[6]	Device discovery Group Formation Power Saving	evaluation (detailed)	wpa_supplicant NS3	Quantified the delay incurred during device discovery and group formation schemes (standard, autonomous and persistent)
[24]	Extension	Multi-hop Wi-Fi Direct network	NS3	Proposed content sharing using multi-hop Wi-Fi Direct networks
[23]	Extension	iDLS over Wi-Fi Direct	ORBIT (testbed)	Proposed Decentralized cOllaborative Media content Streaming (DOMS) using iDLS implemented over Wi-Fi Direct
[25]	Extension	ELBM scheme using Wi-Fi Direct	Matlab	Enhanced Leader Based Multicast (ELBM) is proposed which implements multicast transmission over Wi-Fi Direct. networks.
[18]	Group Formation	Intent Value Computation and GO Election	Matlab	Algorithms proposed to compute Intent Value and GO election.
[14]	Group Formation	MANET and Wi-Fi Direct Cooperation	Android API	Proposed MANET and Wi-Fi Direct internetworking
[16]	Group formation	Criteria for GO election is proposed	Android API	Proposed message delivery over the App. (Mobile Screenshot)
[17]	Group Formation	Clustering algorithm template	Android API	Proposed a template to define new clustering algorithms
[19]	Group Formation	Emergency Group Owner	Testbed	Proposed EGO to maintain near-continuous connectivity of P2P group if GO leaves the group.
[29]	Power Saving	Dynamically Synchronized Power Management (DSPM)	NS3	Proposed DSPM which synchronize the transmission time slots with active time slots in OppPS and NoA schemes.
[22]	Power Saving	Adjustment of awake intervals	NS3	Proposed scheme adjusts the length and number awake intervals according to the PDF of video frames.
[20]	Power Saving	TAPS scheme	NS2	Tunning of parameters in OppPS and NoA schemes for improved performance
[13]	Power Saving	ASPP and AMPP schemes	OPNET Modeler	ASPP used with OppPS and NoA by defining a single presence period in a beacon interval. AMPP used with NoA to define multiple presence periods based on raw bandwidth estimation on the up-link external network.
[21]	Power Saving	DPS Scheme	NS3	DPS scheme is proposed in which the P2P GO toggle between OppPS and NoA schemes based on traffic characteristics. OppPS suits for bulk data transmissions while NoA suits streaming applications.
[26]	Performance	Evaluation (detailed)	Android, iperf	RTT comparison in Wi-Fi Direct Vs. Bluetooth
[27]	Performance	ERiCC-W system using Wi-Fi Direct	Android API	Compare the overhead and delay associated with Wi-Fi Direct and Bluetooth connection in Electrical Business Card Communication over Wi-Fi Direct (EriCC-W) system.
[28]	Group Formation	evaluation (brief)	Matlab	Proposed an election criteria for E-commerce (advertisements dissemination to many users with less complexity)
[7]	Device Discovery	Listen Channel Randomization (LCR)	NS3	New scheme (LCR) proposed to reduce device discovery delay in Find phase

5.1. File Transfer

Transferring files using the Send service in Wi-Fi Direct is simple and efficient as compared to other short range (SR) communication options such as Bluetooth [30]. Several applications are developed on Android OS that make use of Wi-Fi Direct to transfer files between mobiles at Wi-Fi rates. Wi-Fi Shoot, SuperBeam, WiFi Direct Share and SHAREit are examples of file transfer mobile applications available on the Android application store[11].

In [31] authors proposed a platform to build ad-hoc Delay Tolerant Network (DTN) using Wi-Fi Direct. DTN allows source nodes to cache and send data when the connection is established with the destination node. The performance was compared with Bluetooth in Spiderweb. Wi-Fi Direct shows better performance in terms of Device Discovery delay, especially in large networks. A simple chat application was also developed with Wi-Fi Direct.

5.2. Sharing Resources

Wi-Fi Direct can be used to easily print files from smartphones, tablets or laptops directly without connecting to the network. Internet connection sharing was already possible in legacy Wi-Fi using the ad-hoc mode. However, using Wi-Fi Direct the user can scale and extend the network to multi-hops. [32] proposed a tunneling mechanism in Wi-Fi Direct network to allow communication between P2P Groups, thus enabling multi-hop transmission using the Concurrent P2P Device mode. The application was tested using non-rooted Android phones.

5.3. Sharing Applications

Wi-Fi Direct enables users to display photos and videos on multiple screens. In [33] authors designed an application platform called "P2PSIP UA" to share content in the network. The application uses Wi-Fi Direct for creating and managing network connection using Group Formation procedure.

5.4. Streaming Media

Wi-Fi Direct can be used in streaming applications where users can stream contents from streaming video server without connecting to the Internet. Users can form a P2P Group with another user who is connected to the Internet. Thus, the user accesses the server through the P2P GO in the P2P Group. In [34] authors presented a P2P application for live streaming of video contents by using Wi-Fi Direct connections.

5.5. Online Gaming

Users can play multi-player games using Wi-Fi Direct when they make a P2P Group. Higher data rates of Wi-Fi make Wi-Fi Direct an excellent choice for such applications. In [35] authors designed an Augmented Reality 3D Ping Pong game system (ARPP). ARPP use Wi-Fi Direct to search and automatically detect peers.

5.6. Emergency Communication

Wi-Fi Direct can provide easy and flexible connection establishment during emergency communication for alert dissemination [36] when there is no existing infrastructure for communication. Mobile applications like Firechat [11] available on android, can be used to sent messages in event of emergency.

5.7. E-Commerce

Wi-Fi Direct based applications [28] can be designed for sending advertisements locally. For example, shops can send local and targeted promotion messages, when a user is in proximity, in order to gain attention and thus increasing sales value.

5.8. Social Networking

Wi-Fi Direct Device and Service Discovery features allows Wi-Fi Direct to be used for social networking applications like proximity based chatting applications. Another possible usecase is the integration of Internet based social platforms with Wi-Fi Direct based proximity access. In [37] authors proposed a Wi-Fi Direct based P2P social networking framework that enables data exchange between users in closed proximity without using Internet. The prototype is implemented in Android. In [27], Electrical Business Card Communication-Wi-Fi Direct (EriCC-W) is proposed to identify target clients using a gesture of shaking recognized by accelerometer in smartphones.

From the above-mentioned review of recent researches focusing on the realization of different applications based on the Wi-Fi Direct, it can be found that Wi-Fi Direct is being accepted as a potential technology for device-to-device communication.

6. Group Owner Selection

The existing methods for selecting P2P GO based on the Intent Value parameter pose several limitations and drawbacks. In standard Group Formation, only two devices can participate in the GO selection process. In autonomous Group Formation, a device becomes GO without any indication of its actual capabilities to serve as GO. In persistent Group Formation, a P2P device previously serving as GO becomes GO in the re-initiated P2P group without verifying that the capabilities of the P2P GO are not diminished. Due to such limitations, it is of key importance to design a mechanism for selecting the P2P GO in Wi-Fi Direct by revisiting the literature in this field.

We define *selection* as the process where the candidate node does not participate actively in the process or does not announce its desire of becoming the leader of other nodes. Instead, the leader is selected either randomly or based on some selection criteria. Example of such mechanism includes LEACH [41], HEED [42] and SEP [43]. In contrast, the *election* process is the one in which the candidate nodes actively participate in the election process and announce their desire or strength in the network.

In Wi-Fi Direct Standard Group Formation, the P2P GO is elected using GO Negotiation between two P2P Devices. The priority of a node to be elected as P2P GO is measured by its GO Intent value, which is randomly chosen (The Wi-Fi P2P

Table 2: Development in Wi-Fi Direct Applications

Ref	Authors	Year	Proposed Scheme	Applications	Implementation
[34]	Andrea et al.	2013	ICN based P2P Application	Live Streaming	Android
[26]	Je et al.	2014	Large scale multimedia delivery	Live Streaming	Android
[33]	Tuan et al.	2012	P2PSIP Protocol	Content Delivery	Android
[32]	Duan et al.	2014	Multi group communication	Content Delivery	Android
[35]	Xin et al.	2014	Augmented Reality 3D Ping Pong Game (ARPP)	Online Gaming	Android
[38]	Choi and Yun	2014	Wi-Fi Direct for medical application	Medical	Android
[37]	Wang et al.	2014	Mobile Social Networking in Proximity (MSNP)	Social Networking	Android
[27]	Ryo and Yoshito	2014	EriCC-W	Social Networking	Android
[31]	Le and Hitoshi	2014	Sharing Over Delay Tolerant Ad-Hoc Networks (SODAN)	Delay Tolerant Networking (DTN)	Android
[28]	Kim and Lee	2015	Voluntary Advertisement Dissemination Algorithm (VADA)	E-Commerce (Advertisement)	Matlab
[39]	Ahmed and Younis	2015	Alert Dissemination using Service Discovery	Alert Dissemination	Android
[36]	Rajavelsamy et. Al	2016	LTE - Wi-Fi Direct Internetworking	Emergency Communication	Theoretical Aspects
[40]	Luis et. Al	2016	Oi	Opportunistic Data transmission	Android

Table 3: Wi-Fi Direct Based Android Mobile Applications

Application	File transfer	Print	Streaming	Social	Gaming
WiFi Shoot	✓	✓	✗	✗	✗
SuperBeam	✓	✗	✗	✗	✗
Hifi	✓	✗	✗	✗	✗
HitcherChat	✓	✗	✓	✓	✗
Hike Direct	✓	✗	✗	✓	✗
WiFi Direct +	✓	✗	✗	✗	✗
HP ePrint	✗	✓	✗	✗	✗
Wi-Fi Direct Chat APK	✗	✗	✗	✓	✗
FireChat	✓	✗	✗	✓	✗
Wi-Fi Direct Group Chat	✗	✗	✗	✓	✗
WiFi Direct Share	✓	✗	✗	✗	✗
SHAREit	✓	✗	✗	✗	✗
Xbox One	✗	✗	✗	✗	✓

specification does not define any criteria to compute GO Intent value by P2P Devices). The GO Negotiation process in Wi-Fi Direct can be transformed into an election process, where more than one P2P Device can participate, with equal chances, to become P2P GO. [18] The GO Intent value is calculated based on device capabilities e.g. Signal to Noise Ratio (SNR), battery status, processing power, mobility pattern, and number of neighbors (or node's degree).

Relay selection is another research topic in wireless cooperative networks which is an identical process to GO selection. Some reference works on relay selection are given in [44, 45, 46, 47, 48] A relay is defined as an intermediate device between the source and destination nodes. The relay device is responsible to cooperate in the communication between the two devices. There can be single or multiple relays in a network. Networks that implement relay nodes are called multi-hop networks, in contrast to single hop networks in which all nodes communicate directly. Wi-Fi and Cellular networks are examples of single hop networks. Wi-Fi Direct based multi-hop mobile networks can be designed using GO device as relay. Several leader election or relay selection schemes, in Wireless Sensor Networks (WSN) and cooperative networks, can be adopted to elect/select the GO.

We investigated several protocols and algorithms originally proposed for WSN and Cooperative relaying to understand the criteria for P2P GO selection in Wi-Fi Direct. These algorithms are not designed specifically for Wi-Fi Direct and hence may not be integrated in Wi-Fi Direct in their original form. However, some aspects of these schemes can be borrowed for P2P GO selection such as criteria/priority rules for selection and the selection algorithm itself. Following, we discuss briefly these algorithms with reasoning to its applicability into Wi-Fi Direct GO selection problem.

In [49], lowest ID based leader selection is presented with applications specific to Vehicular Ad-hoc networks (VANETs). Each node broadcast its ID to other members in its vicinity. The node with lowest ID is selected as cluster head (CH). The scheme uses stochastic model to reduce the time slots for electing a CH. The lowest ID based algorithms can be modified and adapted in Wi-Fi Direct such as all P2P devices announce their Intent Values (node IDs) using P2P beacon or probe request frames. The device with highest Intent Value (instead of lowest) is selected as P2P GO. The selection algorithm can pose severe delay in larger networks.

In [41], Low Energy Adaptive Clustering Hierarchy (LEACH) protocol is proposed to dynamically select cluster heads based on the nodes' initial energy. The idea of considering device battery status in computing Intent Value for P2P GO has been discussed in [18]. The LEACH algorithm can not be adapted directly for selecting P2P GO because in LEACH, the selected cluster heads are rotated in each round to evenly distribute energy nodes. Whereas in Wi-Fi Direct, the P2P GO is fixed in a given P2P GO. However the idea of alternating roles of P2P GO between two or more devices in a P2P Group can be significant in mobile networks with frequent topology changes.

In [42], Hybrid Energy Efficient Distributed Clustering

(HEED) protocol is proposed to reduce messages overhead and to achieve uniform CH distribution Wireless Sensor Networks (WSN). HEED used two parameters for cluster head selection: a primary parameter called "residual energy" and a secondary parameter called "node's degree". The CH selection scheme is similar to LEACH [41] and hence not suitable for Wi-Fi Direct. However the idea of combined metric for computing GO Intent Value can be readily adapted into Wi-Fi Direct.

In [43], Stable Election Protocol (SEP) is proposed to highlight some of the shortcomings of LEACH protocol. SEP is based on heterogeneous nodes, in terms of their initial energy and location, and provides longer stability period and a better throughput. SEP uses the same selection algorithm as in LEACH [41] and HEED [42].

In [50], an Enhanced-RSSI scheme is proposed for CH selection in peer to peer networks and sensor networks. E-RSSI scheme ensures low node's energy consumption to prolong network lifetime.

In [51], authors proposed GO selection based on received signal strength indicator (RSSI). The RSSI levels are weighted to map with the Intent Value and node with highest Intent Value is selected as GO. The decision is taken at the application layer, hence no modification at MAC layer is required.

RSSI or signal strength based metrics can lead to increased throughput, however does not guarantee the stability of the network in mobile networks. For more stable and long-lasting networks, the mobility of devices should be taken into account while selecting the P2P GO. In [52], Mobility Based Clustering Algorithm (MOBIC) is proposed for MANETs. It considers node's mobility to select the CH to achieve more stable network. In MOBIC each node calculates its *relative mobility* with respect to other neighboring node using power level of the received *hello* message. The relative mobility is only an indication if the node is moving towards or away from a neighboring node. From the relative mobility calculated for each neighbor, each node calculate its *aggregate mobility*. If a node's *aggregate mobility* is lowest than all its neighbors, it declares itself as cluster head; otherwise, it declares itself as cluster member. The algorithm was evaluated in ns network simulator with mobile ad-hoc networks and . MANETs and Wi-Fi Direct both use the same IEEE physical and medium access mechanisms (IEEE 802.11), hence the MOBIC mobility based metric can be easily used in Wi-Fi Direct. MOBIC based GO selection can be useful to improve life of a P2P group if the devices are subject to frequent local mobility.

MOBIC [52] does not take into account the speed of the mobile devices. The authors in [53] extends the concept by proposing a Mobility Aware (MA) CH selection scheme which is based on the speed difference in order to create stable clusters.

Although RSSI is a good indicator of the signal strength of received packet, however it is not the optimum indicator due to interference in the wireless channel and noise can lead to incorrect indication of the received signal power. Instead of RSSI, Signal to Noise Ratio (SNR) is a more reliable metric as it take into account the noise which cause the signal degradation. In [54], an Optimal Timer based relay selection scheme is presented. The scheme maps the instantaneous SNR of the node to a Timer value to maximize the chances of selecting the best relay node and also minimize the average selection time. The timer based scheme can be utilized to form P2P groups for short sessions.

In [55], a CH selection scheme based on weighted parameters (node's degree, remaining battery, mobility factor and received power level) is proposed. The combined metric in this approach is perfect for computation of Intent Values in diverse applications, however the weight of each parameter shall alter from application to application.

In [56], a position based clustering scheme is proposed. The cluster size is controlled by the distance between CH and its members. The eligibility of CH is derived from the travel time and speed deviation of the node. The position based scheme is applicable to scenarios where the P2P GO is required to stay longer or almost fixed in a given geographical location.

In [57], an Optimal Relay Selection (ORS-MAC) scheme is proposed for cooperative media access control in WLAN. The selection is based on the throughput that a node can provide in cooperative mode. This kind of selection criteria is more suitable for high-throughput application such as video and content delivery.

In [58], a method for dynamic CH selection is proposed based on the perceived probability and survival time. The scheme is suitable for group formation intended for longer sessions.

In [59], the proposed relay selection scheme selects the relay node as the one that maximize the SNR between source and destination to improve the network performance. This kind of selection criteria can be used in Wi-Fi Direct based tethering and application where the P2P GO is used only as a relay between two devices.

To estimate the expected delivery of transmission through P2P GO, the outage probability should be analyzed. Outage probability is calculated based on the SNR values for each receiving packet. In [60], a relay selection scheme is proposed to reduce the probability of outage in fading wireless channels. The scheme uses Channel State Information (CSI) parameter to select the best relay node.

A review of above-mentioned methods and techniques for the selection of an optimum node as a leader or as a relay node, provides a useful insight of GO selection problem. Table 4 provides a summary of leader election schemes discussed in this section. Any of these methods can be applied in Wi-Fi Direct for the selection of P2P GO to achieve the desired performance in a specific application.

After the criteria or parameter for selecting the P2P GO is defined, it need to be mapped with the Intent Value to conform with the Wi-FI Direct standard [1]. Based on the Intent Value, the GO is selected. In the next section, we present our proposed method to select the P2P GO based on randomly chosen Intent Values in Wi-Fi Direct. The proposed method is designed such that it reduces the group formation time considerably.

Table 4: Leader Election Schemes

Ref	Proposed Scheme	Technique	Criteria	Performance Metric	Evaluation Method	Applications
[49]	Lowest ID	Analytical	Average time slots for CH selection	No. of Time slots	Matlab	VANET
[41]	LEACH	Analytical	Remaining energy	Energy consumption	Matlab	WSN
[42]	HEED	Analytical	Residual energy and nodes degree	Message overhead, CH distribution	Matlab	WSN
[43]	SEP	Analytical	Nodes initial energy	Stability Throughput	Matlab	WSN
[50]	E-RSSI	Simulation	Remaining battery, RSSI	Alive nodes, remaining battery, No. of CHs, distance between nodes and CH	SONIR (Matlab based Simulator)	Peer to Peer Networks
[51]	Optimized GO Selection	Simulation	RSSI	Bit rate and No. of devices	Not specified	Wi-Fi Direct
[52]	MOBIC	Simulation	Aggregate relative mobility	Throughput delay	NS2	MANETs
[53]	MA Clustering	Simulation	Relative distance and speed w.r.t neighbors	Average Cluster Head Lifetime (ACHL), No. of Cluster	NS2	VANETs
[54]	Optimum Timer Scheme	Stochastic	Timer based on arbitrary local parameter	Probability of Success of selecting best node	Monti Carlo Simulation	Cooperative Relaying in MANETs
[55]	Weighted Clustering Algorithm	Simulation	Nodes degree, RSSI, Remaining Battery and Stationarity factor	Average number of clusters versus (1) Transmission range, (2) No. of Nodes (3) Speed of Node	OMNET++	MANET
[56]	PPC	Simulation	Nodes priority computed from travel time and speed deviation	Throughput Cluster Radius Cluster- Reconfiguration rate	CORSIM and NS2	VANET
[57]	ORS-MAC	Simulation	Maximum throughput in cooperation mode	Saturation throughput Probability of error	NS2	WLAN Cooperative Networks
[58]	DCHSM	Voronoi diagram Perceived probability	Survival time estimation	Energy consumption Network Lifetime	Matlab	WSN
[59]	Best Relay selection	PDF, CDF, MGF	Maximum SNR	Symbol Error Rate (SER)	Matlab	WSN
[61]	Relay Selection	MRC, MGF, PDF	Transmission rate, Channel state information, Length of data packets	Symbol Error Rate (SER)	Matlab	WLAN
[60]	Optimal Relay Scheme	CDF McLaren series Monti Carlo sim.	Channel State Information	Outage probability	Matlab	Cooperative networks

7. Modified Group Formation

We propose a *Modified Group Formation* scheme in which all the required information to establish a P2P Group during the GO Negotiation procedure are inserted in the Probe Request and Probe Response frame. These frames are sent by devices during Device Discovery as discussed in Section 2.2. Importantly each P2P device inserts its Intent value and the list of discovered devices (and their corresponding Intent values) in the P2P Information Element (IE) attributes within each Probe Request and Probe Response frames. Thus when a device receives the Probe Request or the Probe Response frame from a second device, it can easily determine which of the device is more capable of being a GO without a need for a GO negotiation. The device with the highest GO Intent value can start an autonomous Group Formation and invite all discovered devices to join its group.

The proposed method has the advantage that it is **backward compatible** with P2P devices that use the standard P2P Group Formation only. Using this modified method, the delay caused by GO Negotiation stage is eliminated, thus reducing the overall delay of the Group Formation. The proposed scheme has another advantage: it supports simultaneous Group Formation among more than two P2P devices.

We evaluated the proposed Group Formation method using an Ubuntu-server virtual machine with several virtual wireless interfaces. Each virtual wireless interface is attached to a different P2P node. We created the virtual network interfaces using mac80211_hwsim. Throughout our experiments, we assume that all devices have identical capabilities (but different Intent values) and we do not take into account the improvement/drop of the per-device throughput/battery. In addition, we presume that all devices are discoverable by each other. Furthermore, in order to automatize the test execution, we always pre-provision devices with a Wi-Fi Protected Setup (WPS) PIN.

In the first step, we evaluated the delay associated with each phase of the standard Group Formation scheme (i.e. device discovery, GO negotiation stages). Figure 5 shows the delays associated when two P2P devices try to form a P2P Group as normalized Cumulative Distribution Function (CDF). In average, the device discovery time requires 1070ms. The time elapsed to negotiate the GO and form a group can vary from 850ms to 9000ms and is in average equal to 2198ms (the median is equal to 1958ms). The time required to form a group once the GO negotiation is finished successfully, is in average equal to 903ms (the median is equal to 873ms). As depicted in Figure 5, devices spent more than 50% of the time during the GO negotiation phase once they discovered another device.

In the second step, We evaluated our proposed Modified Group Formation scheme with different number of devices: two devices, and five devices.

Figure 6 and Figure 7 shows the CDF of the Group Formation time for two and five devices respectively. The comparison results between the conventional Group Formation and the modified Group Formation scheme confirm that the proposed method is faster than the conventional P2P Group formation between two devices. The average P2P Group Formation delay is improved by 20% when the proposed P2P Group formation procedure is used. In the second scenario (five devices), the elapsed time to form a P2P Group is equal in average to 8000ms. The modified scheme is almost three times faster than the conventional Wi-Fi Direct P2P Group formation procedure, as measured in [62] and [63].

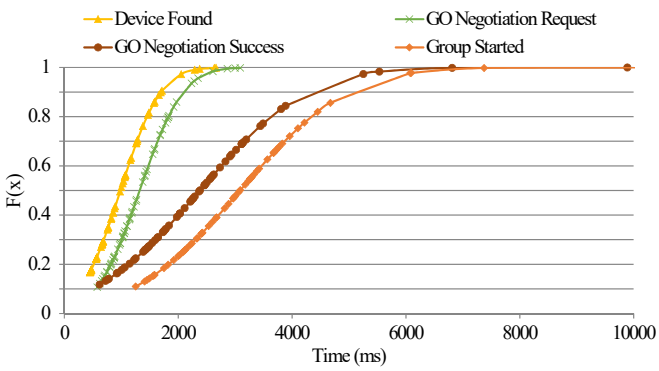

Figure 5: Group Formation Phases

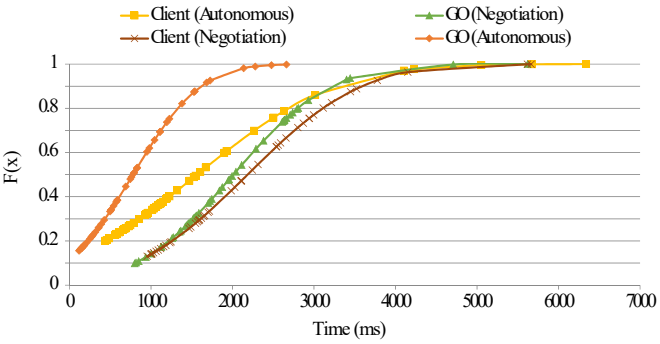

Figure 6: Group Formation Time (two devices)

7.1. Application to the CoCoDi Project

The CoCoDi[12] research project aims at designing, implementing, and optimizing a novel scalable framework for cooperative multimedia multicasting in dense environments with a large number of smartphone users located in a given geographical area. This is particularly applicable to high density network scenarios such as delivering near real-time multimedia content to participants attending an event in a large auditorium. The designed algorithms for the CoCoDi are based on long range multicasting from access points to cluster heads over Wi-Fi 802.11 protocols and short range multicasting within clusters using techniques based on Wi-Fi Direct protocol.

In the context of CoCoDi project, the P2P grouping of mobile stations within a limited area is based on the proposed P2P Group Formation. The purpose is to accelerate the conventional P2P Group Formation method, and make the GO negotiation

[12] http://feaweb.aub.edu.lb/research/d-cocodi/index.html

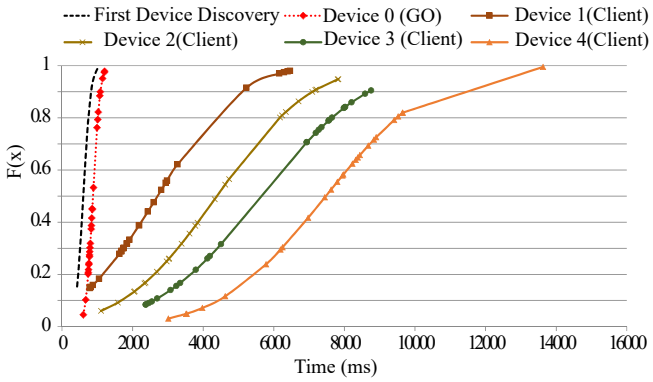

Figure 7: Group Formation Time (five devices)

possible for more than two devices. In addition, the proposed P2P grouping can handle the dynamic peers (moving users) within a premises.

8. Challenges and opportunities

Wi-Fi Direct is primarily designed to enable device to device communication for short range (SR) communication without any existing infrastructure. Several applications are discussed in Section 5. Most of the Android phones comes with built-in support for Wi-Fi Direct. In this section, we discuss several issues and challenges of Wi-Fi Direct implementation and applications. We also highlight the potentials of Wi-Fi Direct and comparison with other SR technologies.

8.1. Energy Efficiency

Wi-Fi Direct devices will more likely run on batteries and therefore energy efficiency is an important factor. Although the protocol includes two new power saving schemes (i.e. OppPS and NoA) for P2P GO as discussed in Section 2.5, energy efficiency is still a challenge in Wi-Fi Direct. The OppPS scheme can save energy only in low traffic conditions whereas the NoA scheme can be implemented to save energy in various traffic conditions. However, NoA scheme require the computation of the optimum length of absence periods. Larger duration of absence periods will save more energy, but will result in less throughput and vice versa.

8.2. Clustering and GO selection

Wi-Fi Direct defines the Group Formation procedure to be used between two P2P Devices. The Wi-Fi Direct specification does not define the mechanism when multiple P2P Devices simultaneously start Device Discovery and Group Formation. As per the current Group Formation procedure, such situation will result in a formation of several P2P Groups, with most of the groups consisting of two devices. This is not desired in several applications like CCN applications. The Group Formation procedure in Wi-Fi Direct can be transformed into an election process where all interested devices form a single large group. An enhanced Group Formation procedure is proposed in [18], where multiple devices collectively elect a GO based on the devices GO Intent values. The GO Intent value was computed using device capabilities to ensure that the most capable device is elected as GO, and thus, improve the group throughput and lifetime. Once the election procedure is realized, the next step is to define whether one or more groups should be formed to achieve the optimum performance. Dividing a large set of nodes into smaller groups is called *clustering*. Clustering techniques are broadly investigated in [49] [64] [65].

8.3. Multi-hop and Scalability

Wi-Fi Direct currently supports single-hop communication. Using P2P Concurrent mode, a P2P Client in one P2P Group can serve as a P2P GO in another P2P Group. An example of such topology is a laptop being a P2P Client in one P2P Group and simultaneously connected to a printer in another P2P Group. The P2P Concurrent Device can be used as a relay node by becoming a GO for another set of devices to fully form multi-hop network. This relay node will be responsible to receive and forward packets between the AP and the P2P Clients. Such operation is not specified in the Wi-Fi P2P specification and should be addressed in the enhancement of the protocol. Such kind of operation will increase scalability of the Wi-Fi Direct network.

8.4. Load balancing for GO

When defining cluster size, the important parameter to consider is that how many clients can be better served by the P2P GO. Even if the best GO is selected, the characteristics of the traffic sent by devices cannot be estimated in several applications, hence we need some kind of load balancing mechanism to implement on P2P GO in order to increase the throughput and reduce the end-to-end delay. Examples of load balancing techniques are given in [66] and [67].

8.5. Copping User's mobility

The current applications of Wi-Fi Direct are only limited to static environments, where users have very limited movement. However, random mobility of users can lead to formation of very unstable P2P Groups. The connection breaking and Group Formation delays might not be desired in some application. Therefore, the mobility of both P2P GO and P2P Clients is a challenging problem. Mobility parameter can be taken into account during the GO election, in order to avoid tearing down of the P2P Group when P2P devices move. However, in several multi-hop scenarios, the mobility can be opportunistically used to provide services. For instance the dissemination of local marketing advertisements [28] can be extended to exploit user's mobility to spread the number of messages send to the target users. Similar applications can be found in alert dissemination as in [39].

8.6. Competition with other SR technologies

Wi-Fi Direct is a Short Range (SR) Communication technology and is based on Wireless LAN standard, the IEEE 802.11,

with enhanced support for D2D communication. Several comparisons of Wireless LANs (IEEE 802.11) with other short-range communication technologies such as Bluetooth (IEEE 802.15.1)[30], ZigBee (IEEE 802.15.4)[68], Dedicated Short-Range Communication (DSRC) [69] and Ultra-Wide Band or UWB (IEEE 802.15.3) are presented in the literature [70, 71, 72, 73, 74, 75].

The common factors for comparison in these studies are spectrum utilization, data rate, energy efficiency, topology, channel access, service discovery and security. From these studies, one can conclude that Wireless LANs and consequently Wi-Fi Direct has several shortcomings as compared to the others SR technologies listed above such as high energy consumption in active/idle mode and expensive hardware chip-sets. The high-energy consumption limits the use of Wi-Fi Direct technology in low power wireless sensors. In contrast Bluetooth and its successor version Bluetooth Low Energy (BLE) [76] provide significant level of energy efficiency and thus these can be better choices for low power sensors, followed by ZigBee. However, Wi-Fi Direct inherits all the advantages of Wireless LANs to compete in the area of D2D connectivity. The potential benefits of Wi-Fi Direct are extremely high data rates ($\sim 250Mbps$), larger coverage ($\sim 100m$), sophisticated security features (WPA-2 using AES 256-bit encryption) and standardized device and service discovery features. These features enable Wi-Fi Direct to become a choice for applications that require high data rates. For example, the backbone or gateway nodes in a wireless sensor networks can be deployed using Wi-Fi Direct.

Apart from the above-mentioned benefits and limitations, Wi-Fi Direct standard includes some potential features which are not yet evaluated. (1) The service discovery feature can be used to automate the group formation and thus networks can be self-organized based on the required services; however, the service discovery procedure is optional and not yet evaluated in the literature. (2) Scalability and realization of multi-hop networks are two of the most important features which can boost Wi-Fi Direct deployments in the future. In terms of scalability, Wi-Fi Direct provides great flexibility. The P2P concurrent mode allows a P2P GO to connect two P2P groups which can be exploited to scale singe-hop Wi-Fi Direct networks or form multi-hop networks [18]. However, the extent to which the Wi-Fi Direct networks can be scaled is also not evaluated in the literature. (3) Majority applications based on short range direct connections will require quick establishment of network connections. The current standard of Wi-Fi Direct poses significant delays in establishing connections as depicted in [6]. The delay associated with device discovery and group formation in case of large-scale networks can be a challenge which shall be quantified and further reduced. (4) The standard offers power saving schemes to save energy for P2P GO, while the P2P clients can also save energy using sleep or doze mode. However, to compete in low energy applications, these power saving schemes need to be further optimized and improved.

9. Conclusion

In this paper, we presented a technical overview of Wi-Fi Direct technology, its network architecture, node's operation such as the Devices and Service Discovery, Group Formation, power saving mechanisms and security features. In Section 3 an overview of implementations of Wi-Fi Direct in network simulators is provided. For the moment, most of the research in the area is performed using Android API and only few of them used network simulators to evaluate the protocol's performance. The lack of simulations based performance study is due to limited or no support of Wi-Fi Direct in all well-known network simulators e.g. NS3, OMNET++, and Riverbed Modeler. It is also possible that individual works cited in this paper have performed partial implementations of Wi-Fi Direct. However, partial implementation should be justified by strong logical arguments and that ignoring some features would not affect the acquired results.

A review of recent research on Wi-Fi Direct was also presented in Section 4. Most of the research work is focused on the performance evaluation of Wi-Fi Direct in several applications. The key research issues highlighted in the literature are energy efficiency, Device Discovery, Group Formation techniques, P2P selection and load balancing and support for multi-hop communication.

In Section 7, we proposed an enhanced scheme for Group Formation in Wi-Fi Direct which is backward compatible with conventional P2P Group Formation. Using the proposed scheme, the Group Formation delay can be reduced by one third of the conventional scheme given in [62] and [63]. In Section 5, a list of Wi-Fi Direct based mobile applications was provided. The section discusses how Wi-Fi Direct has the potential to be used beyond the basic support for device-to-device connectivity and how several applications can be realized using the technology. Issues and challenges related to Wi-Fi Direct are discussed in Section 8.

As future research directions, three key issues of significant importance in Wi-Fi Direct are proposed: (1) reducing *Group Formation* delay, (2) selecting the best Group Owner and (3) support for multi-hop.

Acknowledgment

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